

A Tolerance Is a Request, Not a Measurement

We solve your equation and measure how trustworthy the answer actually is

Luminary Quantum Institute, LLC

We sell the missing half of numerical work: measured truth, not requested tolerance.

Your solver says it converged. That tells you it met the tolerance you asked for. It does not tell you whether the answer is actually trustworthy.

That distinction is the missing half of numerical work.

We solve your equation and measure what the answer is actually worth.

You receive the solution as an explicit function, accuracy measured against the equation's own scale rather than assumed from a requested tolerance, the delivered artifact bound with SHA-256, and a written reproduction scope stating exactly what was checked — and what was not.

Consider two real engagements on a hard plasma-confinement benchmark.

One verified to a relative residual of 3×10^{-10} .

The other did not verify. We delivered it anyway — marked **delivered in full**.

That is not a contradiction. It is the point.

The problem

$$y'' = \mu \sinh(\mu y), \quad y(0) = 0, \quad y(1) = 1$$

Troesch's problem. It is a standard benchmark because it punishes solvers. As μ grows, the solution hugs zero across almost the entire interval and then climbs to 1 inside a boundary layer that thins exponentially. At $\mu = 15$ the initial slope is about 2×10^{-6} . At $\mu = 20$ it is

about 1.7×10^{-8} — the solution leaves the origin almost perfectly flat, and everything happens at the very end.

Two engagements, same equation, two values of μ .

$\mu = 15$ — **VERIFIED**

The acceptance criterion is stated **before** the verdict, not chosen after it: maximum relative residual below 1×10^{-10} of the equation's own scale.

VERIFIED: satisfies the equation to 1.59e-03 on $[0, 1]$ (the equation's scale reaches 2.45e+07; max RELATIVE residual, taken pointwise, is 2.98e-10), and the boundary data to 5.85e-10. Checked by exact coefficient differentiation at interior points, not at the interface nodes. Interface continuity across 34 node(s): max one-sided jump in $(y..y1) = 2.3e-12$ - each side's own coefficients evaluated exactly at the node.

Read the first two numbers together, because either alone would mislead you.

The absolute residual is 1.59×10^{-3} , which sounds poor — until you see that $\sinh(15y)$ drives the equation's own terms to 2.45×10^7 . Measured against the quantities actually being balanced, the pointwise relative residual is 2.98×10^{-10} , roughly three thousand times inside the stated criterion.

We print all three: the defect, the scale it must be judged against, and the result. Quoting only the flattering number is marketing. Quoting only the absolute number is underselling. You should be able to judge.

The seams are measured, not assumed. The solution is delivered as 35 exact power-series pieces. At each of the 34 joins, both sides are evaluated from *their own coefficients* at the node — the largest one-sided jump is 2.3×10^{-12} . And the equation check is performed deliberately *away* from those nodes, where agreement would be uninformative.

An independent check, by a different route. Troesch's equation admits an exact energy relation. We evaluate it at 40-digit precision, with the constant taken from the exact binary value of this run's own initial slope, and measure the drift across 200 samples: 9.6×10^{-5} , on an equation whose terms reach 10^7 . This is not the residual check repeated — it is a conserved quantity that has no reason to hold unless the trajectory is right.

Finding	solved + verified
$y'(0)$	$2.44451303 \times 10^{-6}$
Max relative residual	2.98×10^{-10} (<i>criterion: 10^{-6}</i>)
Boundary data	5.85×10^{-10}
Interface jump, 34 nodes	2.3×10^{-12}
Energy-relation drift	9.6×10^{-5}
Pieces / coverage	35 / [0, 1]
Verification samples	1,387

The report also carries what a vendor would be tempted to delete. One note discloses that some pieces were stretched past their trustworthy range — and explains why that is disclosure rather than contradiction: the substitution check still passed at every sampled point, and the note names the lever that tightens the construction. Another records that an internal notice about a delegated stiff step was **superseded** by a reconciliation check, logged for engineering review rather than quietly dropped.

$\mu = 20$ — NOT VERIFIED, delivered in full

NOT verified: $\max |y^{(2)} - f| = 4.85e+09$ at $x = 1$; max relative residual 1.00e+00. The defect is CONFINED to a BOUNDARY LAYER at $x = 1$ (defect localised within $\sim 1.6e-14$ of the end; 2 verification sample(s) lie inside the layer - it was LOOKED AT, not skipped; interior residual $3.6e-15$): polynomial pieces of this degree cannot resolve a layer this thin - a layer-resolution limit, not an iteration failure. Also checked: boundary data to $7.75e-14$.

This is the deliverable we most want you to read.

The relative residual is **1.00** — the computed y and the right-hand side do not agree *at all* at the worst point. There is no way to soften that, and we do not try. It is the headline of the report.

Then the report does five things a failure message almost never does.

It localizes the failure. The defect is not spread across the interval. It is confined to a boundary layer within 1.6×10^{-14} of $x = 1$. Outside that sliver the **interior residual** is 3.6×10^{-15} . The solution is essentially exact everywhere except a region narrower than a hundredth of a trillionth of the interval.

It proves the layer was not skipped. “2 verification sample(s) lie inside the layer — it was *LOOKED AT*, not skipped.” The easiest way to pass a check is to sample where the trouble isn’t. This report states that it sampled inside its own failure.

It measures where the good region ends. The residual stays under threshold through $x = 0.981009$ and exceeds it beyond — and the report states that any width or location asserted elsewhere in the raw transcript is *superseded* by this measured crossing. Measurement outranks estimate, in writing.

It classifies the cause. “A layer-resolution limit, not an iteration failure.” Those demand opposite responses — re-examine the model, or turn a knob — and the distinction is what separates a useful report from a shrug.

It names the lever. Raise the series degree: `iterations=40`.

The independent energy check agrees, and adds something. Its drift is 2.4×10^8 — a catastrophic failure of a conserved quantity, reached by a completely different route than the residual — and in failing, it locates a pole at $x = 0.99839$. Two independent methods, one verdict, plus a structural explanation of what is there.

Note also that the boundary data on this run is met to 7.75×10^{-14} . Boundary conditions satisfied; equation not verified in the layer. Two different claims, reported separately, neither permitted to stand in for the other.

The engagement is marked DELIVERED IN FULL. Every promised item is present and system-checked. A NOT VERIFIED finding is a *measurement*, not a fulfillment state. It tells you where the problem breaks and which lever to pull. It is never hidden and never softened.

You don’t just get a green light. You get evidence.

A measured result when it passes. A measured failure when it doesn’t. And enough information to understand the difference.

A solver that only tells you when it succeeded is a solver you cannot fully trust when it stays quiet.

Three tiers. One honesty policy.

Tier 1 — Verified. For students and engineers who need a fast, inexpensive check before committing to a formal audit. Ships the artifact, its SHA-256, a branded report in Markdown and PDF, solution and residual figures, and the order manifest — including the finding, the full verify facts, the artifact coverage and the reproduction scope.

Tier 2 — Certified Standard. For industrial and enterprise clients building an auditable paper trail — aerospace, energy, and anyone submitting evidence to a regulator or internal quality function. Adds a formal certificate making exactly four measurable claims: constraint satisfaction at dense off-node points; seam continuity with each side evaluated from its own coefficients; measured tail decay per piece, stated explicitly as an indicator and *not* a continuous-interval remainder bound; and an independent forward check where an invariant exists — and where none exists, the certificate says so, and states that forward error is not bounded by the report while backward error still is.

Tier 3 — Certified Premium. For researchers and advanced engineering teams integrating results into their own workflows or publications. Adds the full certificate with diagnostics, localization and a claims ledger, plus a Premium Addendum: convergence analytics, conditioning indicators showing where the work went, optional perturbation sensitivity, the decomposed recurrence system exported in LaTeX, and further figures including per-component panels, invariant drift and a layer zoom.

Every tier ships the artifact, the hash, and the reproduction scope. **The tiers differ in documentation depth. They never differ in honesty.**

What no tier covers, stated here rather than in a footnote: whether the equation models your system; agreement with another solver, which is a second opinion and not a reproduction; and a solver’s tolerance presented as an accuracy claim. Tier 2 is evidence for a compliance file — it is not compliance itself, and it is not regulatory approval.

Don’t take our word for it

Both engagements are published in full — artifacts, hashes, reports, certificates, figures, manifests.

Every printed number is measured and recomputable from the artifact with Python and `sha256sum`. The artifact carries its own reconstruction rule — $y(x) = \sum c_k(x - x_{left})^k$ on each segment — so the solution can be rebuilt and re-checked without our software. To confirm a file is byte-identical to what we delivered:

```
sha256sum LQI-2026-Troesch-15-T1.artifact.json
```

A line-ending guard and a hash self-check script ship with the deliverables, so the published hashes hold on Windows, macOS and Linux alike. If a hash you compute does not match, tell us — we would rather send you the bytes directly than have you assume the worst.

Download them. Check the hash. Reproduce our measurements — **or reproduce our failure.**

<https://github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library>

Pricing depends on the problem. Send us yours and we'll quote it.

info@mitrainstituteofeducation.org

MIE

GitHub

*SIRPY 0.59.15 · Luminary Quantum Institute, LLC (dba Mitra Institute of Education).
Every number above is measured, recorded in the published artifacts, and recomputable from
them. Client names and order identifiers in the sample manifests are illustrative.*

#SIRPY #NumericalAnalysis #DifferentialEquations #ScientificComputing #Verification
#Reproducibility #ComputationalMathematics #Engineering